

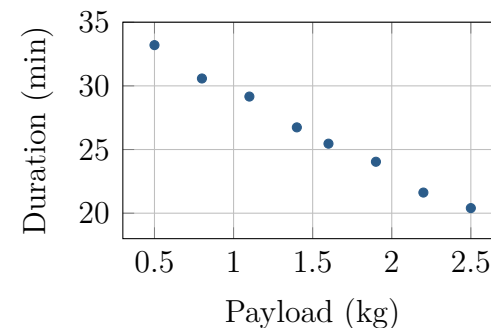
Two Payloads, One Regression Line

Unit 5 · Topic 5.3 · TODAY'S PROBLEM

Suppose eight drone tests with payloads from 0.5 to 2.5 kg produced the shown data and the model $\hat{y} = 36.0 - 6.4x$, where x is payload and $\hat{y}$ is predicted flight duration in minutes. A proposed mission carries 3.5 kg and requires 12 minutes of flight.

Ravi: Approve it: the model predicts more than the required 12 minutes, and the fitted data are close to a line.

Mina: Before deciding, compare the support for predictions at 1.8 and 3.5 kg and identify evidence the safety decision still needs.



FIRST TAKE (5 min, on your sheet)

Would you approve now, delay approval, or approve conditionally? Cite one number or plot feature.

- DOK 1** (a) Identify the explanatory and response variables, slope, intercept, and observed payload range.
- DOK 2** (b) Interpret the slope. Predict duration at 1.8 and 3.5 kg, classify each prediction, and calculate the modeled buffer above 12 minutes.
- DOK 3** (c) Evaluate both approaches using the observed range, the distance from 3.5 kg to that range, the modeled buffer you computed, and the in-range plot. Decide what to do, explain whether the two predictions have equal support, specify evidence needed before approval, and state the consequence of the rejected reasoning.

Video + follow-along: u2_lesson6_live.html (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.

